

Dynamic Robotic Fibers: Liquid Crystal Elastomers for Programmable and Reversible Shape-Changing Behaviors

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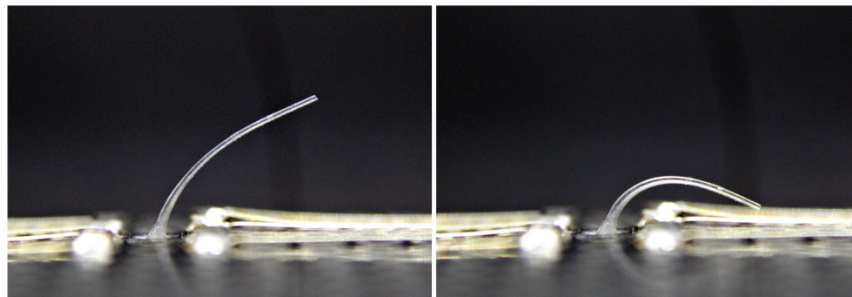


Figure 1: Liquid Crystal Elastomer (LCE) thermo-responsive fiber actuating.

ABSTRACT

This work aims to provide early-stage insights into an electro-thermally actuated liquid crystal elastomer (LCE) fiber for novel shape-changing behaviors that are both programmable and reversible. We build a control system and experimentally investigate the electro-thermal characteristics and actuation, identifying four categories of fiber behavior: Oscillating Tip, Oscillating Fiber, Tilting and Bending. These key parameters illustrate the broad application potential of the proposed approach within a functional, communication and expressive context. Our contributions are three-fold a.) the control of an electro-thermal responsive LCE fiber, b.) directions for soft robotic device integration c.) early-stage insights into fiber shape-changing behaviors towards application.

CCS CONCEPTS

• **CCS Description: Human-centered computing; • Human computer interaction (HCI); • Interactive system and tools;**

KEYWORDS

Liquid Crystal Elastomers, Soft Robotics, Electro-thermal, Interactive Materials, Soft Haptics



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CHI '22 Extended Abstracts, April 29–May 05, 2022, New Orleans, LA, USA
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ACM ISBN 978-1-4503-9156-6/22/04.
<https://doi.org/10.1145/3491101.3519769>

ACM Reference Format:

Amy Winters, Iris Bekkers, Dorsa Nayeb Ghanbar Hosseini, Verindi Vekemans, Samuël Weima, and Miguel Bruns. 2022. Dynamic Robotic Fibers: Liquid Crystal Elastomers for Programmable and Reversible Shape-Changing Behaviors. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29–May 05, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3491101.3519769>

1 INTRODUCTION

At the cross-section of HCI and material science, soft complex systems such as soft robotics and interactive textiles emerge using tightly coupled sensing and actuation, resulting in more advanced user-system interactions. Actuators used in HCI contexts are frequently based on pneumatics or motors, providing a limited range of localized haptic feedback for applications such as simulated touch, and there has been a call for high-resolution haptic and visual output with increased actuation speed for shape-changing interfaces [2]. In contrast, material science advances in soft actuation mechanisms offer diverse structures with complex shape-changing abilities, but perceptual qualities are often overlooked.

A new cross-disciplinary field of Organic Haptics, couples material chemistry, device integration and perception where molecular control of material characteristics are required for both "haptic technologies (functional) and tactile perception (experiential)" (Lipomi et al.) [22]. Here, as proposed by White and Broer [40], Liquid Crystal Elastomers (LCEs), may enable a range of stimuli-responsive shape changing behaviors for haptics. Unlike many other soft (e.g., pneumatic) and hard (e.g., electrical motor) actuating systems, the actuation is incorporated into the molecular structure [7], enabling

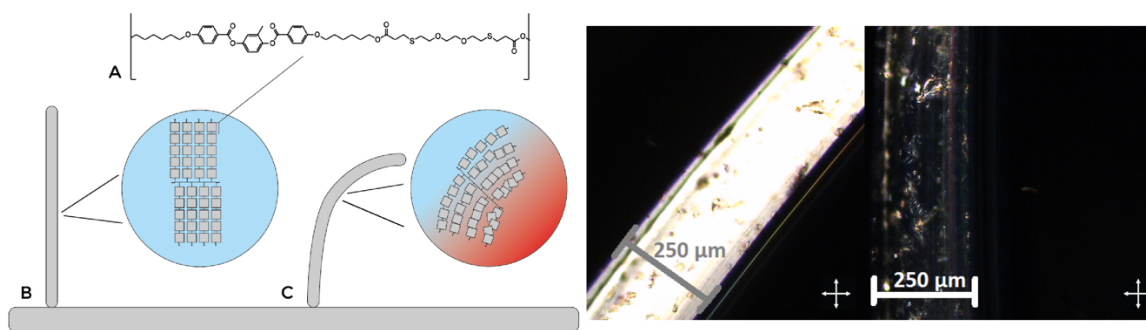


Figure 2: Left. a) Molecular structure of a liquid crystal building block. b) At a uniform and cool temperature, molecular ordering yields a straight fiber. c) Local heating introduces a temperature gradient which disturbs the molecular order and thereby produces macroscopic bending of the fiber. Prior chemical locking-in of the alignment allows the fiber to unbend upon removal of the heat source. Right. Polarizing Optical Micrograph (POM) images of halfway along the length of the fiber.

transition states from static to dynamic. For example, LCEs can provide reversible, low-voltage, programmable complex shape transformations suited for tactile and sensory interfaces in soft robotics and interactive textiles [18, 33, 40, 42].

To this end, we seek to understand the functional and experiential properties of an electro-thermally actuated LCE fiber through a cross-disciplinary study (chemical, mechanical engineering, and interaction design). Based on White and Broer’s thermal-mechanical actuation approach [40], we propose a control system, outline a design space for shape-changing behaviors, and identify four categories of fiber behavior: Oscillating Tip, Oscillating Fiber, Tilting, and Bending. Temperature-triggered behaviors (twisting, swaying, and bouncing) could be examined in the future leveraging timing and control flows, and the control system has the potential to allow programmable customization beyond the four main behaviors identified.

2 RELATED WORK

Advances in material science and soft robotics are laying the foundation for rich, high-definition tactile feedback in the field of haptic technology [3, 22]. However, current approaches hold drawbacks. Readily available actuators are stiff, bulky, and slow, e.g., pneumatics/hydraulics [26, 27, 41] and shape memory alloys (SMAs) [9]. Whilst polymers/electrohydraulics are dangerous at the human interface (2-5kV) [13, 32]. More recent developments in soft systems [11], such as hydrogel actuators [16, 19] or self-healing materials [28], yield mechanical instability, and electrostatic actuators, like pneumatics, are confined to pre-patterned area actuation [21]. Tactile feedback devices require flexibility to interface with our skin, and textiles are not only soft but also perform as individual robotic components (i.e., sensors, actuators, and integration) [33, 36, 42] on both a functional and experiential level. Building on prior work, fiber-driven mechanisms such as OmniFiber, fluidic fiber actuators [1] and ModiFiber, nylon thread actuators [12] are line-based material systems that allow themselves to be structured into textiles (e.g., weaving, knitting, embroidery). Although LCE fibers are still in their infancy, when manufactured as filaments, their yarn-like characteristics may enable more intricate integration into various

textile forms [42]. Further, hair-like surfaces (fur, carpets) offer potential in HCI as soft interfaces for haptic feedback with expressive features [4, 6, 31, 35].

3 LCE FIBER ACTUATION PRINCIPLE

Liquid crystals combine the alignment of crystals with the mobility of liquids. This mobility allows the material to be forced into a desired alignment, which can then be chemically locked, i.e., cross-linked. If suitably designed, the resulting liquid crystal elastomers can be reversibly actuated via heat, light, or magnetism [7, 41]. When drawn from small droplets, these liquid crystal elastomers take the form of fibers. Light and magnetically actuated liquid crystal fibers, utilizing azo compounds and tiny magnetic particles, respectively, have been previously researched and yield appreciable movement [7, 14, 23, 40]. Resistive (Joule) heating presents a viable alternative for applications where magnets are undesirable, and light might not reach. When activated by heat, LCEs generate a molecular-level structural change which results in observable shape-changing behavior (Figure 2). This allows for a locally controllable LCE fiber actuator programmed via a microcontroller. Aside from straight fibers, alternative unactuated forms are also feasible, although fabrication is more complex.

In our study, an upstanding LCE fiber is attached to an electricity-controlled heater. When the LCE is heated locally by connecting a resistor to a power supply, a temperature gradient disturbs the molecular order and produces macroscopic bending of the fiber. Connecting an additional resistor bends the fiber in another direction and ultimately, an array of fibers can be produced. Several parameters can influence actuation behavior, including activation temperature, heat source position, initial curvature, substrate, and fiber shape. Our control system uses a specific heat source configuration to offer early fiber behavior insights, but additional actuation behaviors may be produced by experimenting with fiber geometries, e.g., length and diameter or heat source designs beyond resistors. Microscopic analysis (Figure 2) revealed that the fiber is best aligned halfway along its length which matches with the experimental result that the fiber was best actuated by applying heat at that height.

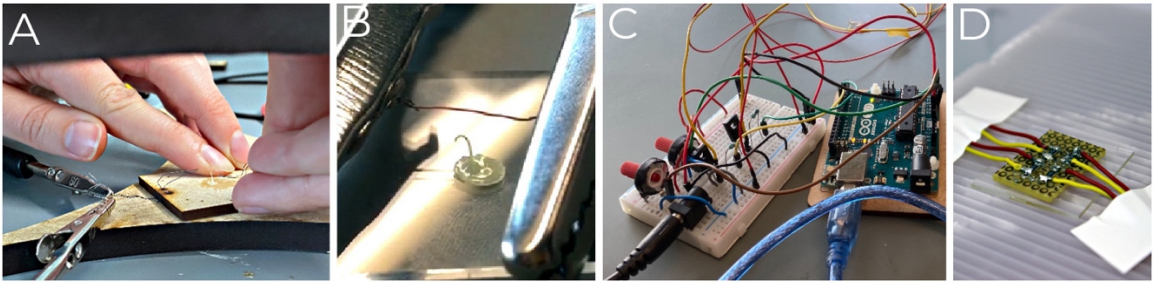
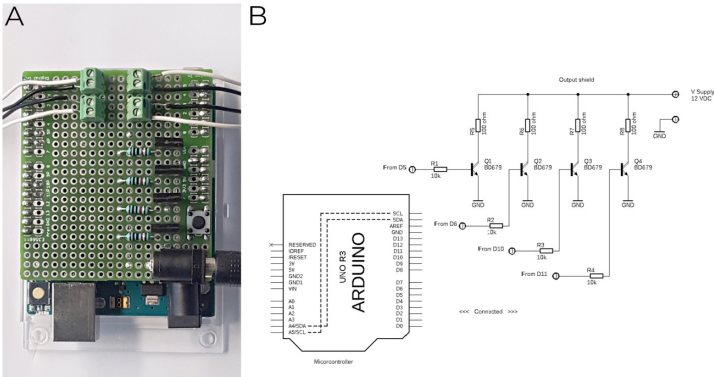


Figure 3: Early Development (a) NiCr wire, MDF substrate (b) NiCr wire, a glass substrate (c) Breadboard prototyping with a potentiometer (d) Square PCB with four 100-ohm resistors.



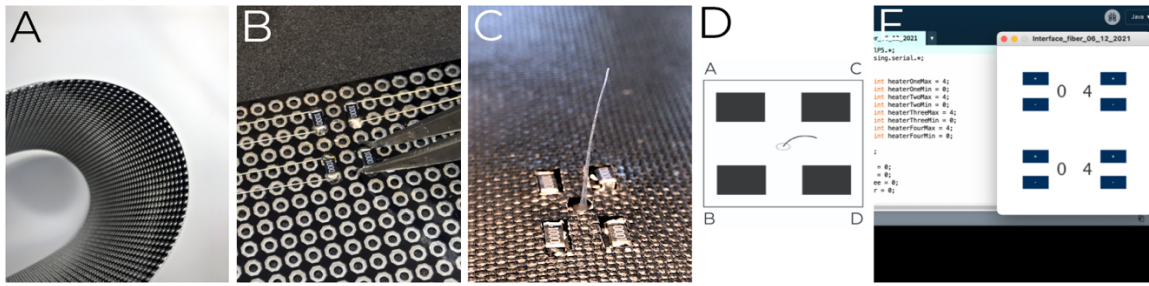


Figure 5: Circuit Fabrication Process. (a) Flexible substrate FR-4 (b) Soldered resistors (c) Optional flexible Teflon cover (d) diagram with designations and the position of the fiber (e) processing interface.

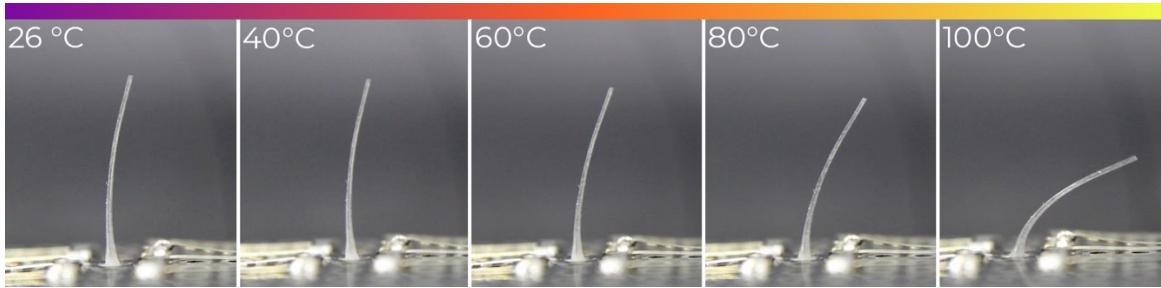


Figure 6: LCE fiber actuation when increasing the temperature with the heat source on side C+D with X2 heated resistors at setting 4 for 120 seconds.

5 ACTUATION PROPERTIES

5.1 Transition Analysis

The relationship between temperature, time and LCE fiber actuation was evaluated using a 14mm length, 0.25mm diameter LCE fiber and our designed control system. Two resistors were heated in position C+D (Figure 5) at processing interface setting 4 (Figure 5, Table 1), and we observed the behavior of the LCE fiber for 120 seconds (recorded by a camera and temperature measured at the base of the fiber with FLIR ONE Thermal Imager). LCE 4D printed elements [25] have shown that progressive heating accelerates actuation. Figure 6 demonstrates the transition movements of the fiber, where we extracted four behaviors (oscillating tip, oscillating fiber, tilting, and bending) and a temperature threshold for each transition: 7 seconds elapsed from applying heat to the very first visible change of oscillation at the tip (34°C). The oscillation moved rapidly from the tip to the entire fiber at 50°C, after 15 seconds and continued to tilt linearly until it reached a full bend deformation at 100°C after 105 seconds. At higher temperatures, over 100°C, the fiber starts to exhibit fast bends or “bouncing” behavior, along with rotational movements. Heat transfer is both time and location-dependent, and therefore exposure to higher temperatures or exposure for a longer period will influence the bending angle. Thus, the resistors may be configured differently and programmed at a higher dissipation rate to accelerate the circuit’s reaction time in future research. The fiber recovery time is approximately 17 seconds from the maximum bent state, which is comparable to Printed Paper Actuator [39] and quicker than SMPs like Foldio (3 minutes) [30]. Heat transmission from the resistors keeps the substrate heated longer, causing the tip

to gently oscillate and straighten during cooling. An active cooling system is required to effectively time the heating and cooling cycles for closed-loop temperature control systems.

5.2 Movement Frequency

Timing and sequencing of transitions are crucial for interaction design applications [9]. To trigger transitional behaviors in the LCE fiber towards more complex expressions for temporal form [38], we can modify the time to transition and effectively “compose” the actuation sequence via the processing interface (Figure 5). Specific behaviors could potentially be designed by maintaining a temperature using PWM in the control system.

We evaluated the fiber using processing settings 1-4 to acquire early insight into fiber behavior at various temperatures across time. The temperature threshold of the fiber behaviors in our movement frequency experiment were comparable to our transition analysis experiment. Time is a key variable in controlling fiber behavior in our control-system configuration, and temperatures sustain specific behaviors for longer periods. For example, setting 1 kept the temperature below 50°C, allowing the fiber to oscillate.

A mechanical engineer and interaction designer observed the movements and qualified the actuation. The mechanical engineer monitored the dynamics, oscillating fiber as “tip going in a circular trajectory” (setting 2), tilting as “curvature and bends” (setting 3) and a full bend where the fiber rotates as “spherical movement on top” (setting 4). The interaction designer recognized experiential characteristics, interpreting oscillating tip as “really soft at top” (setting 1), “tip is waving” (setting 2), oscillating fiber as “whole

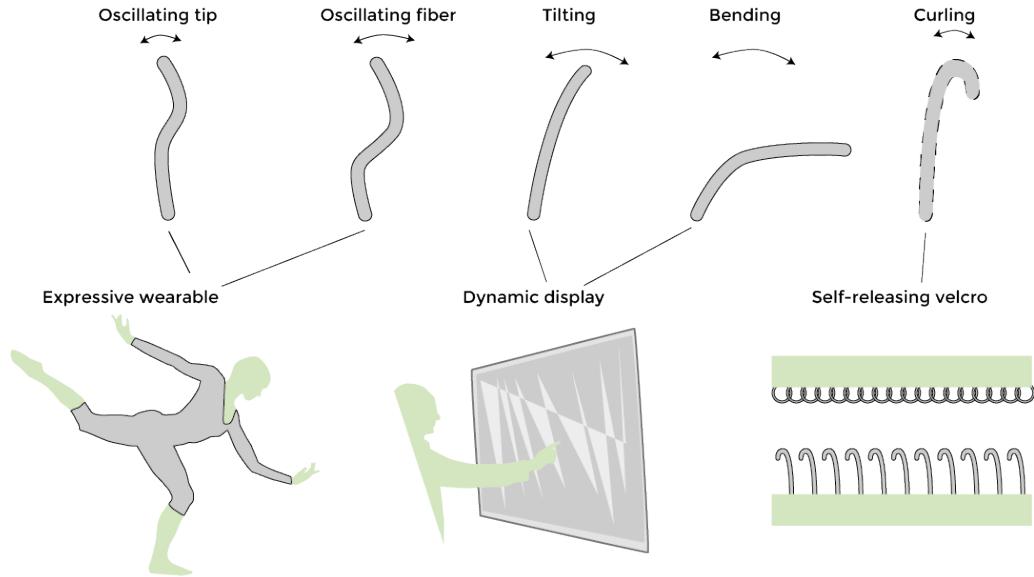


Figure 7: Dynamic Robotic Fibers Design Space.

fiber breathing” and “shaking” (setting 2), slow tilt back and forth as “swaying wind” (setting 3) and a fast bend as a “bounce” (setting 3).

6 APPLICATIONS

The following applications investigate the possibilities of soft haptic devices provided by LCE fiber actuators in expressive, communicative, and functional contexts. While this study focuses on one fiber, a fiber array could provide collective dynamics or propulsion [14]. Future work will develop an actuator array and interactivity through incorporating sensors into the control system (e.g., sound, capacitive).

6.1 Expressive Wearable

The fibers demonstrated oscillating behavior, ranging from oscillating at the tip to an oscillating movement of the entire fiber (base-to-tip) (Figure 7). Sensory stimuli are vital in how we engage with our environment therefore designing an expressive wearable soft interface with a variety of on-skin/on-body haptic sensations has potential. Given that oscillating activity occurs at lower temperatures, this actuation could be used to mimic human touch experiences via soft robotic devices, such as “mediated social touch” (e.g., tickling) [34]. Alternatively, it can be used to send and receive the expressive actions over a distance, such as stroking, tapping, friction or to create an extra dimension of tactility in AR/VR experiences for somatosensory interaction [43] through expressive actions between the user and system [4].

6.2 Dynamic Display

With tilting and bending of the fiber, the potential for sensory data displays is demonstrated, specifically with the collective bending of multiple fibers [14]. Arrays of actuated fibers can act as tactile pixels, transmitting information by optical means and conveying the morphology and touch reaction of diverse surfaces and textiles on-demand (Figure 7). The principle exploited for this concept is the visual difference between flattened and non-flattened fur, an approach used to draw on fabric artefacts (e.g., carpets, garments and plush toys) [35]. Creating a tactile display with LCE fibers would go beyond the static images of Graffiti Fur [35] to allow for a dynamic display with input/output. The bending, tilting and straight fibers can create three different shades on this tactile display. The display can be used as a non-intrusive display on soft surfaces in the house to display data (e.g., energy use), aesthetic shapes or notifications as a form of a qualitative interface [24].

6.3 Self-releasing Velcro

The fiber’s placement-displacement mechanism may transport, grip, or release objects. Our example, self-releasing Velcro, is a soft thermo-responsive switchable adhesive device, an approach towards voltage-switchable surfaces [15]. In conventional hook and loop fasteners, fibers interlock by pressing two sides together, requiring manual pulling to release (Figure 7). The fastener system can respond to heat rather than mechanical force by replacing hooks with LCE fibers. Preliminary tests detected fiber tip curling behaviors (Figure 3b), demonstrating the possibility of building a controlled-release hook and loop fastener. On one side, LCE fibers create small hair-like hooks, and the reverse is made up of smaller

loops made from the same material. Here, in contrast to patented Velcro-like structures (SMAs) [5], LCEs fibers show promise due to their strength and ease of device integration.

7 DISCUSSION, LIMITATIONS AND FUTURE WORK

In this late-breaking work, we aim to offer three contributions, a.) the control of an electro-thermal responsive LCE fiber, b.) directions for soft robotic device integration c.) early-stage insights into fiber shape-changing behavior towards application.

7.1 Control System

We developed a control system for a shape changing LCE fiber and identified transitional temperatures for actuation. Further, a low voltage control system makes the fiber compatible with lightweight programmable microcontrollers. LCE fibers, however, have multiple challenges. One of the limitations of this research is that the behavior of the fiber is not always predictable. Several variables influence the behavior of the fiber. The transition analysis demonstrates that heat conduction from the substrate affects actuation. While the fiber can return to its resting state in 17 seconds, cooling to ambient temperature will take longer. Active heating-cooling (micro-Peltier elements) [29], and microfluidic heating/cooling systems [20] may be one step towards closed-loop temperature control systems.

7.2 Soft Device Integration and Fiber Design

As mentioned in the actuation principle, to actuate fibers from their base, the fiber fabrication process must be adapted to yield fibers well aligned at their base. Once adapted, researchers can expand this early-stage work to printed electronics and substrate resistive heating by starting with heat-resistant flexible materials for soft devices (Teflon, Kapton and fiberglass). At the same time, textile processes (printing, weaving, knitting, and embroidery) could combine the mechanical and expressive properties of actuating fibers with soft sensors [33, 36, 42]. LCEs can undergo complex shape changes based on programmed geometries and molecular orientations [8, 18, 25] and given the LCE geometry affects actuation behavior, fiber shapes could offer tailored behaviors towards applications and novel forms [7, 40], e.g., self-releasing Velcro (Figure 7). In addition to rendering haptic material properties, LCEs can also contribute to the high-resolution visual output of shape-changing interfaces, a challenge addressed by Alexander et al. [2], as the tips of fibers can be composed of thermo-chromic materials.

7.3 Fiber Behavior

We aimed to gauge initial perceptions of the fiber from a mechanical engineer and industrial designer observing the actuation. Both are valuable observations within multidisciplinary research. The interaction designer can provide insight into experiential and expressive features, whereas the mechanical engineer can understand function and reproducibility. Time-based analysis and user research studies with formal evaluations are necessary to assess the entire design space of fiber behavior. We observed other mechanical shape-changing behaviors, such as fiber rotation, which requires a 360-degree camera to precisely capture the direction, bending angles, and rotational movement, together with motion software,

such as Image J as used in Printed Paper Actuator [39] for shape-changing interfaces. The processing interface can help researchers prototype for this temporal dimension, such as time-controlled feedback behaviors (e.g., speed, pace, or direction change), which could be evaluated through a study using temporal form [38]. Further, psychophysical evaluations between the user and actuator [22, 34] could inform device integration towards novel application areas in soft haptics. As a result, building a complete kinetic actuation vocabulary that takes both the perceptual and technical components of the LCE fiber into account would be a promising new research direction.

8 CONCLUSION

This paper introduced Dynamic Robotic Fibers; an electro-thermal actuated LCE fiber that can achieve programmable actuation. We document the control system required to actuate a LCE fiber, a step towards device integration for soft robotics and extract four dynamic shape-changing behaviors whilst outlining potentials for the movement frequency of the fiber and the timing mechanisms required for interaction design prototyping. Here, tactile characteristics can be manipulated at a molecular level in real-time to produce novel tactile and kinesthetic feedback. Through this preliminary research, we aimed to demonstrate the potential of LCE fibers for use in shape-changing interfaces and the cross-disciplinary field of Organic Haptics.

ACKNOWLEDGMENTS

We would like to thank Dick Broer, Danqing Liu, and Li Liu from the Stimuli-responsive Functional Materials and Devices Group (SFD), Chemical Engineering and Chemistry department for supporting with the LCE fiber and research contributions. We would also like to thank Henk Apeldoorn, Geert van den Boomen, Chet Bangaru, Jasper Sterk and Jan Rouvroye from the Industrial Design department for their technical support throughout the project.

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